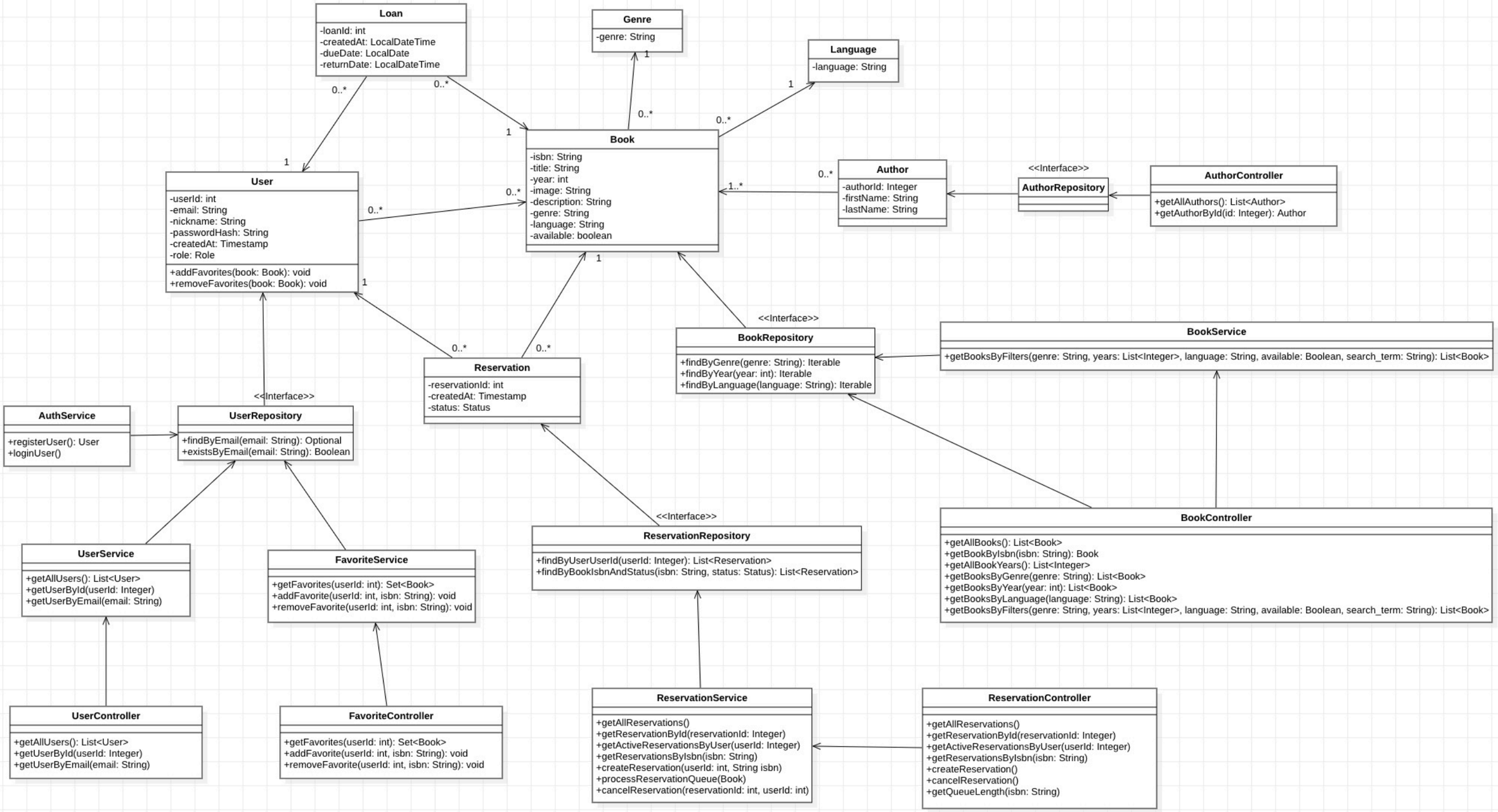

This class diagram describes the backend structure of the Metbook e-library application, built with Spring Boot. This diagram shows domain entities, controllers, services, and repository interfaces. The frontend (React) is excluded from this diagram.

Domain-classes (Entity):
- User - Represents a student or teacher. Users can add or remove books from favorites.
- Book – Represents an e-book that user can reserve and borrow. Only one copy exists per book.
- Author – Represents the writer of a Book. There is many-to-many relationship between Author and Book through the writes join table.
- Reservation – Represents a reservation made by a user when a book that is not available. A user can have multiple reservations, and a book can have multiple reservations forming a queue.

Entity Relationships:
- User and Book have many-to-many relation through favorite join table. User can favorite multiple books, and a book can be favorited by multiple users.
- User and Reservation have a one-to-many relationship, where a user can have multiple reservations.
- Book and Reservation have a one-to-many relation. A book can have multiple reservations forming a queue.
- Book and Author have many to many relations. A book can have multiple authors, and an author can have multiple books.

Controller Classes: Controllers provide REST-API endpoints for the frontend. Each controller handles HTTP requests related to its domain area and delegates business logic to the corresponding Services class.

Service Classes: Services contain the business logic of the application. They process data between controllers and repositories.

Repository Interfaces: Repository interfaces handle database operations. They extend Spring Data's CrudRepository, providing database operations for their entities.

Simplified Classes from the diagram:
- Loan – Same structure as Reservation
- Genre and Language – Similar structures as Author
- Auth - authentication layer, represent the applications authentication

Classes excluded from the diagram:
- NotificationService and MailService – Sends notification to user in email, when returned loan and reserved book available to loan.
- Security layer - This is excluded from the diagram for clarity, as it shows technical implementation classes and do not represent the applications domain.
- DTO classes – Used for data transfer between layers.